# Paul Leger, Ph. D. in Computer Science

Associate Professor & Software Engineer
Civil Engineering School
Catholic University of the North
Larrondo 1281, Coquimbo, Chile

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| Email: | `pleger@gmail.com` |
| Homepage: | `https://pleger.cl` |
| Pragmatics lab: | `https://pragmaticslab.com` (Director) |
| LinkedIn: | `https://cl.linkedin.com/in/plegerm` |


## Personal

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| Name: | Paul Leger |
| Hometown: | Coquimbo, Chile |
| Nationality: | Chilean |
| **Summary:** | |
| Google Scholar: | CrqCdsgAAAAJ |
| ORCID: | 0000-0003-0969-5139 |
| SCOPUS: | 36087550600 |
| DBLP: | https://dblp.org/pid/78/7936.html |
| Directors/Headships: | 6 Positions |
| Indexed Journals: | 37 Publications (32 WoS-JCR, 2 SCOPUS, and 3 Latindex) |
| Indexed Conferences: | 19 Publications (SCOPUS) |
| Indexed Workshops: | 8 Publications (SCOPUS) |
| Journal Editions: | 6 Journals |
| First/Corresp Author: | 24 (represents 37%) |
| Courses: | 56 Courses (49 undergrad and 7 graduate) |
| Thesis Supervisions: | 29 Theses (26 undergraduate and 3 graduate) |
| Relevant Software: | 9 Developments (2 applications, 3 libraries, and 4 languages) |
| External Projects: | 3 with funding |

## Education

**Visiting Researcher**, universities across the world:
- Centre for Development of Advanced Computing (C-DAC), India, 2024 (3 weeks).
- University of The Andes, Colombia, 2021 (3 weeks).
- Tokyo Institute of Technology, Japan, 2019 (6 months).
- University of Cauca, Colombia, 2018 (2 months).
- Shibaura Institute of Technology ($x$ 2), Japan, 2013-2014 (4-2 months).
- École des mines de Nantes, France, 2010 (2 months).

**Certifications:**
-"Fundamentals of English Medium Instruction", University of Pennsylvania, 2023.
-"Teaching University", Highest level, Catholic University of the North, 2024.

**Ph. D. in Computer Science**, University of Chile, 2006 - 2012.
- Subject: Opening Up Trace-Based Mechanisms.
- Advisor: Éric Tanter (https://pleiad.cl/etanter).

**Engineering Degree in Computer Science**, University of La Serena, 1999 - 2005 (July).

**B.Sc. in Computing Science**, University of La Serena, 1999 - 2005 (April).

## Research Keywords

**Programming Languages.** Modularity, abstractions, different approaches, and from the functional programming to novel programming approaches.

**Software Engineering.** Developing tools to help programmers in software construction.

**Agent-Based Models.** The application of agent-based models to predict social behaviors in different environments (*e.g.*, social network sites like Twitter).

## Scientific Services

Manager (*x 11*)

**- Director on the Engineering School at UCN**, Coquimbo (Chile), 2023 - present, https://www.ucn.cl.

**- Pragmatics Lab. Founder and director of the research laboratory in software development and simulations of agent-based models**, Virtual (Chile), 2018 - present, https://pragmaticslab.com.

**- Performance Committee Evaluation for UCN academic units** (**x 2**), Coquimbo (Chile), 2023 - 2024, https://www.ucn.cl.

**- Deputy Director in Engineering Systems Doctorate**, Coquimbo (Chile), 2023 - 2024, https://dsi-en.ucn.cl.

**- Research Secretary at the Engineering School**, Coquimbo (Chile), 2023 - 2024, https://eic.ucn.cl.

**- Ph. D. Coordinator of Engineering Systems Doctorate**, Coquimbo (Chile), 2023, https://dsi-en.ucn.cl.

**- Director(Head) of Civil Engineering in Computer Science at the Engineering School**, Coquimbo (Chile), 2021 - 2022, https://eic.ucn.cl.

**- Director(Head) of Technology Information Engineering at the Engineering School**, Coquimbo (Chile), 2021 - 2022, https://eic.ucn.cl.

**- Director on Chilean Computer Science Society (SCCC)**, Santiago (Chile), 2020 - 2022, https://sccc.cl.

**- Member of two Master Programs at the UCN**, Coquimbo (Chile), 2016 - 2018, https://www.ucn.cl.

Funded Projects (*x 4*)

**- Pluralismo 2023 (ANID-Chilean Government) "Evaluación de Estrategias para Diseminar Noticias Falsas Usando Inteligencia Artificial (Assessment of Strategies to Disseminate Fake News Using Artificial Intelligence)" (PLU230018)**, Santiago (Chile), 2023 - 2024, **NOTE:** Principal Researcher, https://anid.cl/concursos/xv-concurso-de-estudios-sobre-pluralismo-2023.

**- Fondo de Tesis Investigación Científico-Tecnológica Pre y Postgrado: "Evolución del Ecosistema de Haskell y el Uso de Monadas: Un Estudio Exploratorio de Stackage (An Exploratory Studies of the Evolution Haskell Ecosystem and Use of Monads)"**, Coquimbo (Chile), 2021 - 2022, **NOTE:** Principal Researcher, https://www.ucn.cl.

**- Fondef-VIU (ANID – Chile) "Análisis y Valorización de Datos de Comportamiento del Usuario Captados Durante Visitas a Pilotos Inmobiliarios y Utilizando Realidad Virtual (Analysis and Evaluation of User Behavior Data Collected During Visits to Real Estate Pilot Projects and Using Virtual Reality)"**, Santiago (Chile), 2019, **NOTE:** Principal Researcher, https://www.anid.cl/investigacion-aplicada/valorizacion-de-la-investigacion-en-la-universidad-viu.

**- KAKENHI-Project (Japanese government) "Software Engineering and Wireless Sensor Networks" (26330089)**, Tokyo (Japan), 2014 - 2018, **NOTE:** Co-Researcher, https://kaken.nii.ac.jp/search/?kw=26330089.

Organizer (*x 4*)

**- Jornadas Chilenas de la Computación (International Conference Chilean)**, Coquimbo (Chile), 2020, **SCOPUS**, https://jcc2020.cl.

**- Workshop on Context-Oriented Programming and Advanced Modularity (COP) - European Conference on Object-Oriented Programming (ECOOP)**, Virtual Event (Germany), 2020, **SCOPUS**, https://2020.ecoop.org/home/COP-2020.

**- Track on Modularization for Practical Software Engineering (MPSE) - International Conference on Bio-inspired Information and Communications Technologies**, New York (USA), 2015, **SCOPUS**, https://bionetics.org.

**- Track on Modularization for Practical Software Engineering (MPSE) - International Conference on Bio-inspired Information and Communications Technologies**, New York (USA), 2014, https://bionetics.org.

Journal Editor (*x 5*)

**- Associate Editor on Frontiers in Computer Science**, ISSN: 2624-9898, Frontiers, 2025 - present, **JCR-WOS**.

**- Editorial Member on Journal of Innovations in Digital Marketing**, ISSN: 2765-8341, Luminous Insights, 2021 - present.

**- Guest Editor on Information and Software Technology - Special Issue on Visualization applied to Software Engineering**, ISSN: 0950-5849, Elsevier, 2021, **JCR-WOS**.

**- Guest Editor on Frontiers in Computer Science - Special Issue on Artificial Intelligence for Software Engineering: Advances**, Applications, and Implications, ISSN: 2624-9898 - Frontiers, **2025**, **NOTE:** 2025.

**- Guest Editor on International Journal of Software Engineering and Knowledge Engineering**, ISSN: 1793-6403, World Scientific, 2015, **JCR-WOS**.

Journal Reviewer (*x 44*)

**- Artificial Intelligence Review**, ISSN: 1573-7462, Springer Nature, 2026, **JCR-WOS**.

**- Scientific Data**, ISSN: 2052-4463, Nature, 2026, **JCR-WOS**.

**- Journal of Systems and Software**, ISSN: 0164-1212, ELSEVIER, 2025, **JCR-WOS**.

**- Frontiers in Computer Science**, ISSN: 2624-9898, Frontiers, 2024, **JCR-WOS**.

**- Journal of Software: Evolution and Process**, ISSN: 2047-7481, WILEY, 2024, **JCR-WOS**.

**- IEEE Transactions on Reliability**, ISSN: 0018-9529, IEEE, 2024, **JCR-WOS**.

**- IEEE Access** (**x 3**), ISSN: 2169-3536, IEEE, 2022 - 2024, **JCR-WOS**.

**- International Journal of Information Systems and Supply Chain Management**, ISSN: 1935-5726, IGI Global Publishing, 2023, **SCOPUS**.

**- Tsinghua Science and Technology** (**x 2**), ISSN: 1007-0214, SciOpen, 2023, **JCR-WOS**.

**- Proceedings of the Institution of Mechanical Engineers - Part C: Journal of Mechanical Engineering Science** (**x 3**), ISSN: 0954-4062, Sage, 2022 - 2023, **JCR-WOS**.

**- Applied Sciences** (**x 3**), ISSN: 2076-3417, MDPI, 2022 - 2023, **JCR-WOS**.

**- Security and Communication Networks**, ISSN: 1939-0122, Hindawi, 2022, **JCR-WOS**.

**- Informatics for Health and Social Care**, ISSN: 1753-8157, IEEE, 2022, **JCR-WOS**.

**- Complexity** (**x 2**), ISSN: 1076-2787, Hindawi, 2022, **JCR-WOS**.

**- Transactions on Consumer Electronics**, ISSN: 0098-3063, IEEE, 2022, **JCR-WOS**.

**- Education Research International**, ISSN: 2090-4002, Hindawi, 2022, **SCOPUS**.

**- International Journal of Computer Games Technology**, ISSN: 1687-7047, Hindawi, 2022, **SCOPUS**.

**- Journal of Simulation** (**x 2**), ISSN: 1747-7778, Taylor and Francis, 2019 - 2022, **JCR-WOS**.

**- IEEE Latin America Transactions** (**x 8**), ISSN: 1548-0992, IEEE, 2018 - 2022, **JCR-WOS**.

**- Ingeniería y Competitividad**, ISSN: 0123-3033, University of the Valley, 2021, **SCIELO-COLOMBIA**.

**- Journal of Computer Languages** (**x 2**), ISSN: 2590-1184, Elsevier, 2020, **JCR-WOS**.

**- Soft Computing**, ISSN: 1432-7643, Springer, 2019, **JCR-WOS**.

**- Ingeniare** (**x 2**), ISSN: 0718-3305, Universidad de Tarapacá, 2018, **SCOPUS**.

**- Formación Universitaria**, ISSN: 0718-5006, CIT, 2018, **SCOPUS**.

**- Spanish Journal of Marketing**, ISSN: 2444-9709, Emerald, 2017, **SCOPUS**.

**- Journal of Technology Management & Innovation**, ISSN: 0718-2724, Universidad Alberto Hurtado, 2015, **SCOPUS**.

Chair (*x 9*)

**- Industry Track on IEEE International Conference on Software Maintenance and Evolution (ICSME)**, Bogota (Colombia), 2023, **SCOPUS**, **NOTE:** Core-Rank A, https://conf.researchr.org/home/icsme-2023.

**- Lo Mejor de lo Nuestro (LMN) ("the best of ourselves")**, Santiago (Chile), 2022, **SCOPUS**, https://jcc2022.ing.puc.cl/lmn2022.

**- International Conference of the Chilean Computer Science Society (SCCC)**, Coquimbo (Chile), 2021, **SCOPUS**, **NOTE:** Flagship national CS conference (Chile), https://jcc2021.cl.

**- Context-Oriented Programming and Advanced Modularity (COP)**, Virtual Event (Germany), 2020, **SCOPUS**, https://2020.ecoop.org.

**- Session Chair on Track on Software Engineering (SE)-ACM Symposium on Applied Computing (SAC)**, Marrakech (Morocco), 2017, **SCOPUS**, https://selab.uos.ac.kr/sacse17.

**- Track on Modularization for Practical Software Engineering (MPSE) - International Conference on Bio-inspired Information and Communications Technologies** (**x 3**), Boston (USA), 2014 - 2016, **SCOPUS**, https://bionetics.org.

**- Publicity Chair on 9th International Conference on Bio-inspired Information and Communications Technologies**, New York (United States), 2015, **SCOPUS**, https://bionetics.org.

Program Committee Member (*x 22*)

**- Chilean Workshop on Quantum Computing and Quantum Software Engineering (QSQCE-Chile)**, Chile, 2026, **SCOPUS**, https://jcc2026.utalca.cl/wsqsqce-en.html.

**- International Conference on Software Technologies (ICSOFT)**, Porto (Portugal), 2026, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

**- (SPLASH) 2025 Workshop on Virtual Machines and Language Implementations**, Singapore, 2025, **SCOPUS**, https://conf.researchr.org/home/icfp-splash-2025/vmil-2025.

**- International Conference on Software Maintenance and Evolution (ICSME) - Industry Track**, Auckland (New Zealand), 2025, **SCOPUS**, https://conf.researchr.org/track/icsme-2025/icsme-2025-industry-track.

**- XXXVI International Marketing Congress (AEMARK)**, Murcia (Spain), 2025, https://aemarkcongresos.com/areas-tematicas-y-comite-cientifico.

**- International Conference on Software Technologies (ICSOFT)**, Bilbao (Spain), 2025, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

**- International Conference on Software Technologies (ICSOFT)**, Dijon (France), 2024, **SCOPUS**, https://icsoft.scitevents.org/ProgramCommittee.aspx.

**- Brazilian Symposium on Programming Languages (SBLP)**, Mato Grosso do Sul (Brazil), 2023, **SCOPUS**, https://cbsoft2023.ufms.br/en-US/sblp.

**- Brazilian Symposium on Programming Languages (SBLP)**, Virtual Event (Brazil), 2022, **SCOPUS**, https://cbsoft2022.facom.ufu.br/sblp.php.

**- Context-Oriented Programming and Advanced Modularity (COP)**, Berlin(Germany), 2022, **SCOPUS**, https://2022.ecoop.org/home/COP-2022.

**- Brazilian Symposium on Programming Languages (SBLP)**, Natal (Brazil), 2020, **SCOPUS**, https://cbsoft2020.imd.ufrn.br/sblp.php.

**- Congreso Internacional de Computación e Informática del Norte de Chile (INFONOR)** (**x 6**), Norte de Chile (Chile), 2013 - 2019, https://infonorchile.cl.

**- ACM Symposium on Applied Computing (SAC) - Track on Software Engineering (SE)**, Pau (France), 2018, **SCOPUS**, https://www.sigapp.org/sac/sac2018.

**- ACM Symposium on Applied Computing (SAC) - Track on Software Engineering (SE)**, Marrakech (Morocco), 2017, **SCOPUS**, https://www.sigapp.org/sac/sac2017.

**- ACM Symposium on Applied Computing (SAC) - Track on Software Engineering (SE)**, Pisa (Italy), 2016, **SCOPUS**, https://www.sigapp.org/sac/sac2016.

**- ACM Symposium on Applied Computing (SAC) - Track on Software Engineering (SE)**, Salamanca (Spain), 2015, **SCOPUS**, https://www.sigapp.org/sac/sac2015.

**- New Ideas or Emerging Results (NIER) and Tool Demo (TD) - Third IEEE Working Conference on Software Visualization (VISSOFT)**, Santiago (Chile), 2015, https://vissoft.dcc.uchile.cl.

Evaluator (*x 16*)

**- Evaluation committee of FONDEF-IT (Health-IT)**, ANID (Chile), 2026, https://anid.cl/concursos/concurso-de-investigacion-tecnologica-idea-2026/.

**- President of the Ph.D thesis commission: "Diversidad de Licencias en Software Libre"**, Universidad Rey Juan Carlos (Spain), 2025, https://en.urjc.es.

**- President of the Ph.D thesis commission: "Integrated Mathematical Models for Agricultural Decision-Making Under Uncertainty: Efficiency; Water Optimization and Resilience in Olive Production."**, Universidad Católica del Norte (Chile), 2025, https://www.ucn.cl.

**- Evaluator of Fondef-IDeA Projects** (**x 4**), ANID (Chile), 2023 - 2025, https://anid.cl/concursos/concurso-idea-id-tecnologias-avanzadas-2025/2025.

**- External Reviewer of Master Thesis: "Keen: Kotlin Genetic Algorithms Framework"**, University of Chile (Chile), 2024, https://www.uchile.cl.

**- Evaluator of Exploration Projects**, ANID (Chile), 2022, https://www.anid.cl/proyectos-de-investigacion/proyectos-de-exploracion.

**- External Reviewer of Master Thesis: "Points-to Analysis for Context-Oriented JavaScript Programs"**, University of the Andes (Colombia), 2022, https://uniandes.edu.co/en.

**- External Reviewer of Master Thesis: "Ad Hoc Systems Management and Specification with Distributed Petri Nets"**, University of the Andes (Colombia), 2021, https://uniandes.edu.co/en.

**- External Reviewer of Master Thesis: "Analysis of WebRTC signaling"**, University of the Andes (Colombia), 2021, https://uniandes.edu.co/en.

**- Research Project**, Pontificia Universidad Católica de Valparaíso (Chile), 2018, https://www.pucv.cl.

**- External Reviewer of Master Thesis: "Verificación progresiva de programas en Dafny"**, Pontificia Universidad Católica de Valparaíso (Chile), 2018, https://pucv.cl.

**- External Reviewer of Master Thesis: "Discovering Memory Optimization Opportunities by Analyzing Shareable Objects"**, University of Chile (Chile), 2017, https://www.uchile.cl.

**- Evaluator on Short Internship Projects (MEC)**, ANID (Chile), 2014, https://www.conicyt.cl/pai.

# Academic Experience

## *Lecturer*

Postgraduate ($x$ 7)

**Applied Research Practice**, Doctorate of Engineering System, Catholic University of the North, 2024.

**Thesis Proposal**, Master Program, Catholic University of the North, 2019 - 2021.

**Advanced Topics in Management of Information Technology II**, Master Program, Catholic University of the North, 2020.

**Advanced Topics in Management of Information Technology I**, Master Program, Catholic University of the North, 2019.

**Research Methodology**, Master Program, Catholic University of the North, 2018.

**Design Patterns**, Computer and Telecommunication Engineering, Diego Portales University, 2011.

**Objects and Aspects**, Computer and Telecommunication Engineering, Diego Portales University, 2010.

Undergraduate ($x$ 48)

**Programming Languages (*x* 8**), Computer Civil Engineering, Catholic University of the North, 2021 - 2025.

**Complex Engineering Problems (*x* 3**), Information and Information Technology, Catholic University of the North, 2024 - 2025.

**Data Structure (*x* 5**), Computer Civil Engineering, Catholic University of the North, 2018 - 2021.

**Agent-Based Model for Social Simulation (Guided Research)**, Computer Civil Engineering, Catholic University of the North, 2021.

**Algorithms and Programming**, Engineering, Catholic University of the North, 2020.

**Applied Science Project**, Computer Civil Engineering, Catholic University of the North, 2020.

**Data Science Application to the use of Programming Languages (Guided Research)**, Computer Civil Engineering, Catholic University of the North, 2020.

**Introduction to Research in Computer Science**, Computer Civil Engineering, Catholic University of the North, 2019.

**Communication Elements (*x* 5**), Information and Management Control, Catholic University of the North, 2014 - 2018.

**Management Information System Development**, Information and Management Control, Catholic University of the North, 2018.

**DataBase (*x* 4**), Information and Management Control, Catholic University of the North, 2013 - 2017.

**Process Modelling (*x* 3**), Information and Management Control, Catholic University of the North, 2013 - 2016.

**Programming (*x* 3**), Information and Management Control, Catholic University of the North, 2013 - 2016.

**Programming (*x* 5**), Industrial Civil Engineering, Diego Portales University, 2010 - 2012.

**Programming and Database**, Business Management, Catholic University of the North, 2012.

**Programming Languages and Paradigms**, Computer Civil Engineer, University of Talca, 2012.

**Advanced Programming**, Industrial Civil Engineering, Diego Portales University, 2010.

**Computing 1 (*x* 3**), Computer and Telecommunication Engineering, Diego Portales University, 2009.

Thesis Advising ($x$ 27)

**Postgraduate (*x* 3)**, Ph. D and Master, Catholic University of the North, 2017 - 2025.

**Undergrad (*x* 24)**, Engineering, Catholic University of the North, 2013 - 2025.

### *Evaluations as Professor*

**Catholic University of the North** (Three consecutive times) Best professor evaluation for undergrad teaching in School of Business Studies, 2012 - 2014.

**Diego Portales University** (Spanish) "Paul dictó varios cursos (incluso cursos de postgrado) en nuestra carrera, él mostró cercanía con los alumnos y capacidad suficiente para motivarlos a interiorizarse en la materia", Jonathan Frez (jonathan.frez@udp.cl), Teaching Management, of School of Computer Science and Telecommunications Engineer telecommunications.

**University of Talca** (Spanish) "Vistos los resultados de la encuesta, el prof. Leger obtiene puntaje por sobre el promedio del Departamento y de la Facultad en cada uno de los ítems de la encuesta...". Ruth Garrido (rgarrido@utalca.cl), Head of School of Computer Science Engineer.

## Publications

My research profile IDs are *ORCID* (0000-0003-0969-5139), *Web of Science* (Q-6174-2017), and *Scopus* (36087550600).

**NOTE**: For conferences/workshops, I use the CORE ranking, which belongs **CO**mputing **RE**search and Education Association of Austral-asia (near to the publication year). CORE is available on http://portal.core.edu.au/conf-ranks.

### *Journals (Indexed)*

Canquil, Diego, Daniel SanMartín, Irene Inoquio-Renteria, and Paul Leger (Jan. 2026). “A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption”. In: *Water Resources Management* 40.2, p. 83. DOI: 10.1007/s11269-026-04496-4. **JCR-WOS**.

Leger, Paul, David Moreno-Lumbreras, Felipe Ruiz, Nicolás Sepúlveda, Ismael Figueroa, and Nicolás Cardozo (Feb. 2026). “The Stackage Repository: An Exploratory Study of its Evolution”. In: *Journal of Computer Languages* 86, p. 101386. DOI: 10.1016/j.cola.2026.101386. **JCR-WOS**.

Leger, Paul, Daniel San-Martín, Diego Nuñez, and Tomás Vélez (Feb. 2026). “An Agent-Based Model Reflective Architecture for Social Network Sites”. In: *Science of Computer Programming* 252, p. 103452. DOI: 10.1016/j.scico.2026.103452. **JCR-WOS**.

Abdi, Farshid, Shaghayegh Abolmakarem, Amir Karbassi-Yazdi, Paul Leger, Yong Tan, and Giuliani Coluccio (Jan. 2025). “Predicting Patients’ Revisit Intention Based on Satisfaction Scores: Combination of Penalized Regression and Neural Networks”. In: *IEEE ACCESS* 13, pp. 2783–2800. DOI: 10.1109/ACCESS.2024.3522767. **JCR-WOS**.

Karbassi, Amir, Yang Tan, and Paul Leger (Jan. 2025). “Clustering Financial Institutions in Countries Based on a Hybrid Random Forest and Induced Ordered Weighted Averaging”. In: *Journal of Management Analytics*, pp. 1–30. DOI: 10.1080/23270012.2025.2454674. **JCR-WOS**.

Terán, Oswaldo, Paul Leger, and Manuela López (Apr. 2025). “Factors that Drive Market Share and the Oligopolistic Character of Cross-border B2C Ecommerce: An agent-based scenario analysis approach”. In: *Simulation: Transactions of the Society for Modeling and Simulation International* 0.0, pp. 1–24. DOI: 10.1177/00375497241296542. **JCR-WOS**.

Junior, Paulo Nocera Alves, Paul Leger, and Isotilia Costa Melo (Oct. 2024). “Efficiency Analysis of Engineering Classes: A DEA Approach Encompassing Active Learning and Expositive Classes Towards Quality Education”. In: *Environmental Science & Policy* 160, p. 103856. DOI: 10.1016/j.envsci.2024.103856. **JCR-WOS**.

Leger, Paul, Hiroaki Fukuda, Nicolás Cardozo, and Daniel San Mártin (Jan. 2024). “Exploring a Self-replication Algorithm to Flexibly Match Patterns”. In: *IEEE ACCESS* 12, pp. 13553–13570. DOI: 10.1109/ACCESS.2024.3355319. **JCR-WOS**.

B, B. Arinze, DA. Buell, P. Cabezas, JD. Caddell, J. Chase, Y. Chen, E. Chi, SJ. Cosgrove, M. Dabkowski, D. DeVasto, B. Duong, N. Elliot, TS. Ellis, M. Feuer, E. Friess, PB. Gallagher, JF. George, V. Gerard, M. Gertrudix Barrio, G. Getto, G. Giordano, C. Gupta, M. Gupta, V. Gupta, GF. Hayhoe, C. Hidalgo-Alcazar, A. Houser, C. Hubka, KM. Jacobsen, I. Kloo, D. Kong, SJ. Kowalewski, JT. Labriola, A. Lancaster, S. Lang, C. Lauer, J. Lee, Paul Leger, CC. Lewis, C. Liles, and E. Londner (Dec. 2023). "2023 Index IEEE Transactions on Professional Communication Vol. 66". In: *IEEE Transactions on Professional Communication* 66.4, pp. 1–6. DOI: 10.1109/tpc.2023.3338309. **OTHER INDEXES**.

Leger, Paul, Alexandre Bergel, Juan Pablo Sandoval Alcocer, and Leonel Merino (Mar. 2023). "Introduction to Special Issue on Visualization Applied to Software Engineering". In: *Information and Software Technology* 155, p. 107118. DOI: 10.1016/j.infsof.2022.107118. **JCR-WOS. A journal edition**.

Leger, Paul, Nicolás Cardozo, and Hidehiko Masuhara (Apr. 2023a). "An Expressive and Modular Layer Activation Mechanism for Context-Oriented Programming". In: *Information and Software Technology* 156, p. 107132. DOI: 10.1016/j.infsof.2022.107132. **JCR-WOS**.

Leger, Paul, Felipe Ruiz, Nicolás Cardozo, and Hiroaki Fukuda (Feb. 2023). "Benefits, Challenges, and Usability Evaluation of DeloreanJS: A Back-in-Time Debugger for JavaScript". In: *PeerJ Computer Science* 9, e1238. DOI: 10.7717/peerj-cs.1238. **JCR-WOS**.

López, Manuela, Carmen Hidalgo-Alcázar, and Paul Leger (May 2023). "The Effect of Message Repetition on Information Diffusion on Twitter: An Agent-Based Approach". In: *IEEE Transactions on Professional Communication*, pp. 150–169. DOI: 10.1109/TPC.2023.3260449. **JCR-WOS**.

Salinas, Matias, Paul Leger, Hiroaki Fukuda, Nicolás Cardozo, Vannessa Duarte, and Ismael Figueroa (Jan. 2023). "Evaluations of Integrated Programming Environment for First-Year Students in Computer Engineering". In: *Journal of Universal Computer Science* 29.1, pp. 73–98. DOI: 10.3897/jucs.81329. **JCR-WOS**.

Manzano, Carlos, Claudio Meneses, Paul Leger, and Hiroaki Fukuda (Apr. 2022). "An Empirical Evaluation of Supervised Learning Methods for Network Malware Identification Based on Feature Selection". In: *Complexity* 2022.18. DOI: 10.1155/2022/6760920. **JCR-WOS**.

Pizarro, Vicky, Paul Leger, Carmen Hidalgo-Alcázar, and Ismael Figueroa (Jan. 2022). "ABM RoutePlanner: An Agent-Based Model Simulation for Suggesting Preference-Based Routes in Spain". In: *Journal of Simulation* 17.4, pp. 444–461. DOI: 10.1080/17477778.2022.2027826. **JCR-WOS**.

Sosa, Juan Sebastian, Paul Leger, Hiroaki Fukuda, and Nicolás Cardozo (July 2022). "Ad Hoc Systems Management and Specification with Distributed Petri Nets". In: *Journal of Parallel and Distributed Computing* 169, pp. 117–129. DOI: 10.1016/j.jpdc.2022.06.015. **JCR-WOS. NOTE: Selected in "the best of ourselves" on JCC-2022 (main event in Chile)**.

Terán, Oswaldo, Paul Leger, and Manuela López (Mar. 2022). "Modeling and Simulating Chinese Cross-border e-Commerce: An Agent-Based Simulation Approach". In: *Journal of Simulation* 17.6, pp. 1–18. DOI: 10.1080/17477778.2022.2043791. **JCR-WOS**.

Camacho, Marta Cecilia, Francisco Álvarez, César Collazos, Paul Leger, Julián Bermúdez, and Julio Ariel Hurtado (June 2021). "A Collaborative Method for Scoping Software Product Lines: a Case Study in a Small Software Company". In: *Applied Sciences - Basel* 11.15. DOI: 10.3390/app11156820. **JCR-WOS**.

Duarte, Vannessa, Paul Leger, Sergio Contreras, and Hiroaki Fukuda (June 2021). "Using Artificial Neural Network to Detect Fetal Alcohol Spectrum Disorder in Children". In: *Applied Sciences (Basel)* 11.13. DOI: 10.3390/app11135961. **JCR-WOS**.

Figueroa, Ismael, Paul Leger, and Hiroaki Fukuda (Jan. 2021). "Which Monads Haskell Developers Use: An Exploratory Study". In: *Science of Computer Programming* 201, p. 102523. DOI: 10.1016/j.scico.2020.102523. **JCR-WOS**.

Fukuda, Hiroaki, Ryota Gunji, Tadahiro Hasegawa, Paul Leger, and Ismael Figueroa (Feb. 2021). "DSSM: Distributed Streaming data Sharing Manager". In: *Sensors* 21.4, pp. 1–15. DOI: 10.3390/s21041344. **JCR-WOS**.

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### *Workshops (SCOPUS)*

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## *Congresses (Conferences/Workshops not indexed)*

**Note: These papers are just presentations of technical reports or working papers.**

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### *Posters*

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Leger, Paul (Nov. 2012). "Opening Up Trace-Based Mechanisms — Application to Context-Aware Systems". PhD thesis. University of Chile, pp. 1–180. **Published in the University of Chile's system**.

## Software Development Engineer

As I enjoy working on software engineering, I occasionally work as a freelancer doing software development and as a team manager in software projects:

**Macrosoftware.** Consulting software engineer, 2020 - present.

**G-Tech.** Project Manager, a startup mobile application, 2022.

**Autopista Central.** Consulting engineer, 2017.

**Regional Observatory (working market).** Software project manager, implementation of an information system that allows users to analyze working market in a region of Chile, 2016 - 2017.

**Autopista Central.** Consulting engineer, 2011.

**Center for Advanced Research in Arid Zones (CEAZA).** Research assistant, two tasks: a) using MM5 (mesoscale model) to simulate or predict mesoscale atmospheric circulation and b) applying artificial neural networks to group/classify rainfall scenarios (both tasks applied to a limited-area in Chile), 2002 - 2004.

**Viña Ocho Tierras.** Project Manager of Web Site, 2002.

## Software Development for Research

Research-driven software systems and developer tools, built with production-grade engineering practices where applicable (open-source repos, demos, and reusable libraries. Source: https://pleger.cl/software.

**PL/SE research with AI:** Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). https://pleger.github.io/PL-SE-research-with-AI, 2026. [**ARTIFICIAL INTELLIGENCE**].

**SBABM:** Java software to simulate buyers in e-commerce like Amazon, AliExpress, and Wish. An agent-based model to simulate the e-commerce behavior between agents and markets. https://github.com/pragmaticslaboratory/SBABM, 2023. [**APPLICATION**].

**EMAjs:** JavaScript library that implements an Expressive and Modular Activation mechanism for context-oriented programming. https://github.com/pragmaticslaboratory/EMAjs, 2022. [**LANGUAGE**].

**DeloreanJS:** A back-in-time debugger for JavaScript. This software is strongly developed by Felipe Ruiz (felipe.ruiz@alumnos.ucn.cl) and Guillermo Victorero (guillermo.victorero@alumnos.ucn.cl). You can try it on https://pleger.cl/sites/deloreanjs. https://github.com/pragmaticslaboratory/deloreanjs, 2019. [**APPLICATION**].

**AspectScript:** An aspect-oriented extension of JavaScript for expressive AOP, which integrates several state-of-the-art features related to higher-order programming and expressive scoping of aspects. https://pleiad.cl/aspectscript, 2010. [**LANGUAGE**].

**FASOW:** FASOW provides an Open Implementation for Agent-Based Models applied to e-WOM Marketing Campaigns. This software is strongly developed by Diego Nuñez (diego.nunez.trabajo@gmail.com). https://github.com/pragmaticslaboratory/fasow, 2024. [**LIBRARY**].

**Matcher Cell:** A JavaScript library to flexibly match patterns. You can try it on https://pragmaticslaboratory.github.io/matcher-cells-study-cases. https://github.com/pragmaticslaboratory/match-cell-base, 2021. [**LANGUAGE**].

**Sync/CC (proof of concept):** A library to address callback issues on JavaScript. You can try it on https://pleger.cl/synccc. https://github.com/pragmaticslaboratory/synnccc, 2019. [**LIBRARY**].

**ESA-JS:** An implementation of ESA, describing an expressive stateful aspect language. A stateful aspect language supports the definition of monitors to observe and react to a program execution trace. They have numerous applications in domains like error detection, security, and modular definition of crosscutting concerns. 2012. [**LANGUAGE**].

**WeCa:** A practical library that allows for modular and flexible control over causality issues on the Web. In contrast to current proposals, WeCa uses stateful aspects, message-ordering strategies, and vector clocks. WeCa has been used with several practical examples from Web applications. For instance, we analyze the flow of information in these applications like Twitter using WeCa. 2012. [**LIBRARY**].

## Awards & Distinctions

**The Best of Ourselves**, two scientific articles have been selected on "The Best of Ourselves" on Chilean Computer Science Society (Chile), 2019 & 2022.

**Funding for Visiting Researcher**, Alliance of Pacific (AGCI 2018) for a research stay, Colombia, 2018. https://alianzapacifico.net/becas-2.

**Best working paper award**, AEMARK, Spain, 2015. https://www.aemarkcongresos.com/congreso2015/es/premios.

**Selected as the best professor for undergrad teaching**, Business School, Catholic University of the North, Chile, 2014 & 2017.

**Ph. D. funding**, Conicyt, Chile, 2007.

**Ph. D. funding**, Nic, Chile, 2006.

## References

**Hidehiko Masuhara**. Ph. D. in Computer Science. Full Professor & Dean. Department of Mathematical and Computing Science. Tokyo Institute of Technology, Japan (masuhara@acm.org). Webpage: https://prg.is.titech.ac.jp/people/masuhara.

**Éric Tanter**. Ph. D. in Computer Science. Full Professor. Computer Science Department (DCC), University of Chile, Chile (etanter@dcc.uchile.cl). Webpage: https://pleiad.cl/people/etanter.

**Alexandre Bergel**. Ph. D. in Computer Science. Computer Scientist. Rational AI (https://relational.ai), Switzerland (alexandre.bergel@me.com). Webpage: https://bergel.eu.

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